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Equal rates of drug resistance in leprosy cases with relapse and recurrent /chronic type 2 reaction; Time to revise the guidelines for drug resistance testing in leprosy

Running head- Drug resistance in leprosy relapse and reactions

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What is already known about this topic?

- Emerging cases of drug resistant *M. leprae* is a major cause for leprosy relapse
- Drug resistance studies are usually only undertaken in cases of relapse with positive skin smears or new multibacillary leprosy cases.

What does this study add?

- In cases of relapse as well as in those with chronic/recurrent type 2 leprosy reactions; drug resistance was found at almost equal rates to the tune of 16%.
- Drug resistant *M. leprae* was also found in a few cases of skin smear negative relapse.

Abstract

Background- Leprosy relapse/recurrence is a serious concern especially in a leprosy endemic nation like India. It is believed that bacilli persisting even after multi-drug therapy could cause relapse; recently however drug resistance as a cause for recurrences and chronic erythema nodosum leprosum (ENL) has been speculated. Our objective was to study drug resistance patterns in cases of leprosy relapse and chronic/recurrent ENL's.

Methodology- This cross-sectional study conducted over a period of 1 year included patients diagnosed as leprosy relapse and those with chronic/recurrent (c/r) ENL. Skin biopsy specimens were subjected to conventional PCR for resistance testing for rifampicin, dapsone and ofloxacin, targeting *rpoB*, *folP* and *gyrA* gene of *M.leprae* respectively.

Results- A total of 61 patients (25 smear negative) were included in the study. Of these, 37 were diagnosed as leprosy relapse and 24 had c/r ENL. Drug resistance to at least one drug was identified in 10 (16.4%) cases. Rates of drug resistance were 5.4% (2/37) for dapsone, 10.8% (4/37) for rifampicin and 2.7% (1/37) for ofloxacin amongst cases of relapse where as it was

12.5% (3/24) and 8.3% (2/24) for dapsone and rifampicin respectively amongst those with c/rENL.. Multi drug resistance was seen in 3.3% (2/61) patients.

Conclusion - Drug resistance rate amongst those with c/rENL was almost equaled that of relapse. Smear negative leprosy relapse cases also had resistance to bactericidal drugs. These findings call for modifications in criteria for testing under leprosy drug resistance surveillance and all cases of relapse and those with recalcitrant c/rENL should be tested.

Introduction

It was in 2005 that India achieved leprosy elimination, however, with 62% of new cases reported worldwide, leprosy still remains a public health concern in India.¹ In the 1960's, the first case of dapsone resistant *M. leprae* was reported. Due to dapsone monotherapy, the rates of resistance levels rose to as high 40% in few areas. This led to the introduction of multidrug therapy (MDT) for leprosy in early 1980s.² However, since then multiple reports of resistance to rifampicin and ofloxacin have also emerged.³ Primary drug resistance even prior to starting MDT have also been increasingly reported.⁴ In studies from India, the rate of drug resistance in leprosy (relapse and new cases) has ranged from 3.9-8.1%.⁴⁻⁷ More recently, there have been reports of drug resistance as a cause for recalcitrant leprosy lesions and chronic type 2 reactions.^{8,9} Erythema nodosum leprosum (ENL) is the cutaneous manifestation of a type 2 reaction and this could be classified as acute, chronic or recurrent.¹⁰ The objective of this study was to assess the patterns of drug resistance of *M. leprae* in cases of relapse and those with chronic/recurrent type erythema nodosum leprosum (c/rENL).

Methodology

Study design

This cross-sectional observational study was conducted at the leprosy clinic in a tertiary care center in North India between July 2019 to July 2020. All consecutive consenting adults who were either diagnosed as leprosy relapse or who had c/rENL based on the following definitions were included.

Case Definitions:

Relapse was defined as a patient who has completed a full course of MDT for leprosy in the past and who returns with signs and symptoms of the disease that are not deemed to be due to a reaction according to the clinician.¹¹ Those fulfilling this criteria for relapse after having taken appropriate MDT and ensuring compliance to treatment were included.

Recurrent ENL: A second or subsequent episode of ENL occurring 28 days or more after stopping treatment for ENL.¹⁰

Chronic ENL- Presence of ENL for at least 24 weeks or more during which a patient has required ENL treatment either continuously or where any treatment free period has been 27 days or less.¹⁰

In all these patients, slit skin smear (SSS) and 4 mm punch skin biopsy specimens were sent for histopathology and drug resistance testing.

Drug resistance testing-

DNA was extracted from biopsy specimen by using DNeasy Blood and Tissue Kit (Qiagen Franklin Lakes, NJ). qPCR- High Resolution Melting (HRM) analysis for drug susceptibility testing of *M. leprae* was done for rifampicin, dapsone and ofloxacin targeting DRDR (drug resistance determining region) fragments in *rpoB*, *folP* and *gyrA* gene. Conventional PCR was performed as per WHO protocol for all three drugs rifampicin, dapsone and ofloxacin targeting DRDR fragments in *rpoB*, *folP* and *gyrA* gene of *M. leprae* respectively followed by sequencing (Eurofin Lab, Hyderabad, India). Samples were analysed by both HRM and DNA sequencing to avoid false negative reports. Sequence data were analysed by using MEGA 5.1 (<http://megasoftware.net/>) and BLAST.

Based on the results, treatment was modified as per WHO guidelines.¹² Patients with rifampicin resistance were treated using at least two of the following second-line drugs: clarithromycin, minocycline, or a quinolone (ofloxacin, levofloxacin or moxifloxacin), plus clofazimine daily for 6 months, followed by clofazimine plus one of the second-line drugs daily for an additional 18 months. In case of rifampicin plus ofloxacin resistance, the regimen followed was clofazimine, clarithromycin and minocycline for 6 months followed by clarithromycin or minocycline plus clofazimine for an additional 18 months.¹²

Results

A total of 61 patients were included (44 males, 17 females) in this study of which 25 were smear negative. Of these, 37 were diagnosed as leprosy relapse and 24 had c/rENL. Amongst patients with relapse, 6 patients later went on to develop c/r ENL during their disease course. They were calculated under relapse primarily as that was the criteria used to conduct resistance testing in them. Demographic data of these patients has been outlined in table 1. Drug resistance study was conducted for these 61 patients of which 10 (16.4%) were found to have resistance to at least one anti-leprosy drug. Rates of drug resistance was found to be 5.4% (2/37) for dapsone, 10.8% (4/37) for rifampicin and 2.7% (1/37) for ofloxacin amongst cases of relapse. Amongst those with c/rENL, rates of resistance were 12.5% (3/24) and 8.3% (2/24) for dapsone and rifampicin respectively. There was improvement in skin lesions and reactional episodes within 6 months of starting alternate leprosy treatment (ALT) in patients with drug resistance and c/r ENL. The clinical profile and results of drug resistance testing of these 10 patients has been outlined in table 2.

Discussion

Results from a global leprosy survey on antimicrobial resistance conducted between 2009 and 2015 revealed a rate of 8% for overall drug resistance (included both relapse and new cases). Countries with more than 10 cases of rifampicin resistance were Brazil (32) and India (18).⁴ In a multicentric study (2015) from India, resistance to rifampicin was seen in 3.6% (4/111) of the samples and to dapsone and ofloxacin was noted in 8.1% (9/111) each.⁵

The amino acid residues 53 and 55 of the *folP1* gene have been shown to confer dapsone resistance. Missense mutations at codon 55, resulting in substitutions of arginine or leucine for proline are often identified.^{13,14} In our study, dapsone resistance was seen in 5 cases (2 relapse and 3 with c/rENL). In a study from India, the rate of dapsone resistance amongst cases of relapse was 8.1% (9/111).⁵ In another study from Eastern Uttar Pradesh, no dapsone resistance was found in their cohort of 25 patients who underwent *folP1* gene sequencing (17 new cases, 6 on MDT and 2 treatment defaulters).⁶ In our study, all 5 dapsone-resistant cases had the same mutation (Pro55Leu) in the drug resistance-determining region of the *M. leprae folP1* gene. This is in accordance with the above study, wherein majority of dapsone resistance cases were found to have the same mutation (Pro55Leu).⁵ This is one of the most frequent mutations seen in *folP1*.²

Mycobacterial resistance to rifampicin is primarily due to missense mutations within codons of a highly conserved region of the *rpoB* gene.¹⁵ In our study, 6 patients were found to have resistance to rifampicin (4 relapse and 2 r/c ENL). Distinct mutations were detected in these strains (Thr433Ile and Asp441Tyr; Gly432Arg and Gln438His; Ser456Leu). Substitution within codon Ser-456-Leu is the most frequent mutation associated with development of rifampicin-resistance.² As per WHO guidelines published in 2017¹⁰, the mutations at codon positions 441 and 438 have been proven for conferring drug resistance through mouse foot pad experiments. Mutations at codon positions 432 to 437 were mentioned to be reported with caution. Had we found these in isolation, we would have had to confirm this with mouse foot pad method. However since we observed them in combination with mutations at codon position 441 and 438; we reported mutation at position 432 and 437 as resistant. In a study from Eastern India, 2 out 50 cases of relapse were found to have resistance to rifampicin and both cases had the mutation Glu442His in the *rpoB* gene.⁷ But later it was found that this codon position does not confer resistance to rifampicin by mouse foot pad method.¹⁶ In another study from India, rifampicin resistance was seen in 4/111 (3.6%) cases of relapse with two new mutations (Phe439Leu and Gln442His in *rpoB*).⁵ Contrary to our study, the study from eastern Uttar Pradesh found no rifampicin resistance amongst 32 cases sequenced (22 new cases, 6 patients on MDT, 2 defaulters, 2 cases of relapse).⁶ Development of resistance to ofloxacin has been associated with mutations within a highly conserved region of *gyrA*.¹⁵ The most frequent mutation seen is Ala91Val.² In this study, ofloxacin resistance was noted only in case 2 who was smear negative relapse and also had resistance to rifampicin. Detailed report of this case has been published earlier.¹⁷ In a study from India, 2 out 111 cases had multidrug resistance to dapsone and ofloxacin.⁵ In the global survey, cases from Brazil, India and Indonesia were found to have multidrug resistance - dapsone plus rifampicin and dapsone plus ofloxacin. No cases with resistance to rifampicin and ofloxacin were noted.⁴ In present study, two out of the 10 cases had multi-drug resistance. Both these cases are unusual since they could have been missed out if routine drug resistance testing guidelines were to be followed. Case 2 was a smear negative and case 8 had lepromatous leprosy with c/rENL. It was interesting to note that case 4 who also had a smear negative relapse presented with new onset nerve function impairment without skin lesions in whom drug resistance was documented on nerve biopsy specimen. In earlier reports, drug resistance studies were performed

in nerve biopsy specimens in patients with predominantly neural leprosy. Drug resistance was identified in 6/77 patients, including 2 patients with multidrug resistance.¹⁸

Routine drug resistance testing in case of c/rENL is not part of protocols of drug resistance testing in leprosy.¹⁰ In an earlier study from our centre, 16/35 (47.5%) patients amongst those treated with ALT (due to unresponsiveness to WHO MBR-MDT) had c/rENL. On follow up, the number of reactional episodes significantly reduced in all these patients. However drug resistance studies were not done.¹⁹ In another case report from North India, drug resistance testing in a case of lepromatous leprosy with recurrent ENL's revealed rifampicin resistance. ALT was initiated which led to significant improvement in number of reactional episodes.⁸ We observed that, 4/24 (16.6%) cases of ENL's in this study had drug resistance. Multidrug resistance was also noted in one patient. All patients reported reduction in number and severity of reactional episodes on changing to ALT.

Conclusion

Emergence of drug resistance, especially in the post elimination phase is a cause for great concern. We found drug resistance to at least one of the drugs in 16.2% of relapse cases and in 16.6% of c/rENL's. Two patients were found to have multi-drug resistance. Drug resistance studies are generally not done in cases of c/r ENL. This aspect needs further studies, since drug resistance rates amongst those with c/r ENL was almost equal to that amongst relapse. In addition, 3 cases in our cohort had a smear negative relapse (including one pure neural relapse); and all 3 tested positive for resistance to bactericidal drugs. These findings certainly call for modifications in criteria for selection of leprosy patients who are made to undergo drug resistance testing.

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Table 1- Demographic data of study participants

Gender	Male	44 (72.1%)
	Female	17 (27.8%)
Age (year)	Mean	35.3
Type of leprosy (baseline)	Borderline tuberculoid	26 (42.6%)
	Borderline lepromatous	19 (31.1%)
	Lepromatous leprosy	16 (26.2%)
Mean BI (range)		2.42
Smear negative relapse cases		25 (40.9%)
Total Relapse		37 (60.7%)
Relapse with recurrent ENL		6
Recurrent/chronic ENL		24 (39.3%)
Overall drug resistance rate		10/61 (16.4%)
Drug resistance rate amongst relapse		6/37 (16.2%)
Drug resistance rate amongst recurrent/chronic ENL		4/24 (16.6%)
Rate of resistance to dapsone		5/61 (8.2%)
Rate of resistance to rifampicin		6/61 (9.8%)
Rate of resistance to ofloxacin		1/61 (1.6%)
Rate of multi drug resistance		2/61 (3.3%)

Table 2- Clinical profile of patients who tested positive for drug resistance

		6	5	4	3	2	1	Pt. No.
35		61	35	28	61	35	43	Age
Male		Female	Male		Male	Male	Male	Gender
Zirakpur, Punjab	Saharanpur, Uttar Pradesh	Unna, Himachal	Chandigarh	Ambala, Haryana	Chandigarh	Chandigarh	Saharanpur, Uttar Pradesh	State
LL	BL	LL	BT	BT	BT	BT	BT	Initial spectrum at diagnosis
Chronic/Recurrent ENL	Relapse	Relapse-patches with recurrent ENL	Relapse- NFI only, no patches	Relapse-patches	Relapse-patches+NFI	Relapse-patches + NFI	Relapse-patches + NFI	Reason for drug resistance testing
4	4	6	0	0	0	0	4	BI at presentation
Foamy macrophages, neutrophilic infiltrate, Fites strong positive	Foamy macrophages, Fites strong positive	Foamy macrophages, Fites strong positive	Nerve biopsy-	Epithelioid granuloma, Fites negative	Ill defined granuloma, Fites negative	Granuloma with foamy macrophages, Fites positive		Skin Biopsy findings
8 months	Completed 12 months	Completed 12 months	Completed 12 months	Completed 18 months	Completed 12 months	Completed 12 months	Completed 12 months	Duration of taking MBR MDT
Rifampicin	Rifampicin	Dapsone	Rifampicin	Rifampicin	Rifampicin	Rifampicin + Ofloxacin	Dapsone	Drug resistance
Ser456Leu	Gly432Arg; Gln438His	Pro55Leu	Confirmed by HRM analysis only	Confirmed by HRM analysis only	Confirmed by HRM analysis only	RIF-Thr 433 Ile and Asp 441 Tyr Ofloxacin- Gly 89 Cys	Pro55Leu	Mutations
C,O,M-6months, C,M-18 months	C,O,M-6months, C,M-18 months	MBR MDT(without dapsone) x 12 months	C,O,M-6months, C,M-18 months	C,O,M-6months, C,M-18 months	C,C,I,M-6months, C,M-18months	C,C,I,M-6months, C,M-18months	MBRMDTx 12 months	Treatment modification

10	9	8
20	30	34
Female	Male	Male
Fatehgarh sahib, Punjab	Morena, Madhya Pradesh	Barnala, Punjab
LL	BL	LL
Chronic/Recurr ent ENL	Chronic/Recurr ent ENL	Chronic/Recurr ent ENL
3	4	5
Foamy macrophages, neutrophilic infiltrate, Fites strong positive	Foamy macrophages, neutrophilic infiltrate, Fites strong positive	Foamy macrophages, neutrophilic infiltrate, Fites strong positive
Completed 12 months	Completed 12 months	Completed 12 months
Dapsone	Dapsone	Dapsone+ Rifampicin
Pro55Leu	Pro55Leu	Pro55Leu Ser456Leu
Treatment for ENL's	Treatment for ENL's	C,O,M- 6months, C,M- 18 months

Abbreviations- c/rENL- chronic/recurrent erythema nodosum leprosum, RIF- rifampicin, O- ofloxacin, DAP- dapsone, MDT multidrug therapy, MBR- multibacillary, BI- bacteriological index, BT borderline tuberculoid, BL borderline lepromatous, LL lepromatous leprosy, NFI nerve function impairment, C- clofazimine, Cl clarithromycin, M minocycline